

Aimal Asim

Full Stack Developer — Backend & Real-Time Systems

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Self-taught full stack developer (MERN) focused on backend architecture and real-time systems. Built a 340-endpoint learning platform and a load-tested medical exam prep platform, both handling production-scale traffic. No CS degree, high school then a development bootcamp, then straight into shipping production systems.

EXPERIENCE

Full Stack Developer — BlackMode

Aug 2025 – Present

ExamExpert — Learning Management System

- Built the backend and full-stack integration from the ground up, joining an existing frontend prototype and shipping 38 MongoDB models across 340 API endpoints
- Diagnosed a production bottleneck under load and traced it to database connection pool limits; refactored auth middleware and connection handling to take capacity from ~100 concurrent users failing at 60s P95 to 500 concurrent users at 88ms P95 with zero failures, verified with k6
- Implemented real-time features over Socket.IO (live notifications, support ticketing), JWT auth with Google OAuth, brute-force protection, and PayFast payment integration
- Configured SMTP email delivery and DNS records directly, no third-party provider, per infrastructure requirements

AceMrceM — Medical Exam Prep Platform

- Built the backend for a 31-model, 180-endpoint platform, with CI/CD through GitHub Actions after local builds outgrew the production VPS's available memory
- Diagnosed and fixed a false-positive DDoS rule that was blocking legitimate traffic under load; verified with a 15-point security test suite, 15 of 15 checks passed
- Load tested the production build: 80 concurrent virtual users, 190ms P95 response time, 99.8% success rate across 10,764 requests; estimated capacity of 200–400 real concurrent users
- Integrated PayFast for payment processing

SKILLS

Frontend React & TypeScript, Redux, React Query, Tailwind CSS, Framer Motion, responsive design

Backend Node.js & Express, MongoDB & Mongoose, PostgreSQL & SQL, Socket.IO, REST API design, JWT auth

Infrastructure Docker, Git, Linode VPS provisioning, CapRover / Coolify, CI/CD (GitHub Actions), load testing (k6), email/DNS configuration, security hardening

EDUCATION

PSDC Development Bootcamp

High School Diploma